

Some problems on digit representations

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Note: the order of the problems does not unravel anything about their difficulty. In fact, the order is alphabetical. Problems 2, 13, and 14 are borrowed from various sources; the rest of them were proposed by the lecturer for various contests.

1. A mathematician born in the 20th century noted:

My phone number consists of six digits. The three-digit number formed by the first three digits differs from that formed by the last three. The first three digits are nonzero and distinct, and if we consider all the other five three-digit numbers formed by permuting those digits, the sum of these five numbers equals my birth year. The last three digits are also nonzero and distinct, and if we consider all the other five three-digit numbers formed by permuting those digits, the sum of these five gives the birth year of my twin brother.

Was this mathematician born on a weekend?

2. A positive integer n is called an *IMSC number in base b* if it can be obtained by applying the four basic operations, exponentiation, and concatenation, possibly with parentheses, to its own digits in the base b , where each digit is used exactly as many times as it appears in n , and the digits must appear in the same order as they do in n . (For example, 127 is an IMSC number in base 10, since $127 = -1 + 2^7$.)

- a) Find all bases b in which there are infinitely many IMSC numbers.
- b) Find all bases b in which there are infinitely many prime IMSC numbers.

(Clarification: at least one operation that is not concatenation must be applied.)

3. Donald Duck is standing in front of a blackboard on which a positive integer is written, whose two digits have been erased by a sponge, and between them there is an even number of (known) digits. Donald knows the remainders of the initial number upon division by 9 and by 11, and he is trying to deduce the erased digits. It turns out that Donald is unable to fulfill his desire—that is, there are multiple ordered pairs (x, y) such that, for each such pair, after the digits from that pair are written in the corresponding spots, the obtained number upon division by 9 and by 11 gives precisely the remainders known to Donald. Determine which are the possibilities for the ordered pair (x, y) that Donald cannot decide on.
4. Find all positive integers with 31,012,015 digits such that, whenever any of the digits is replaced by another digit, the resulting number is never divisible by 11.
5. Let $c_0 > 0$ and $c_1 \geq 0$ be integers. Prove that for any $K \in \mathbb{N}$ there exists $n \in \mathbb{N}$ and a list of bases $\{b_1, b_2, \dots, b_K\}$ such that:
 - a) for each i such that $1 \leq i \leq K$, n is a 3-digit palindrome in base b_i , and that, for some i_0 such that $1 \leq i_0 \leq K$, the expansion of n in the base b_{i_0} is $\langle c_0, c_1, c_0 \rangle$;
 - b) for each i such that $1 \leq i \leq K$, n is a 4-digit palindrome in base b_i , and that, for some i_0 such that $1 \leq i_0 \leq K$, the expansion of n in the base b_{i_0} is $\langle c_0, c_1, c_1, c_0 \rangle$. ✱
6.
 - a) Let $f(x)$ denote the sum of digits of a positive integer x . Prove that for each positive integer n there exists a positive integer p such that the equation $f(np x) = f(x)$ has a solution x that does not contain the digit 9 among its digits.
 - b) Is it true that for any n we may choose $p = 1$?
7. Let m and n be perfect 17th powers with 15 062 017 and 15 062 017 000 digits, respectively. Can the number obtained by writing m and n one after the other be a perfect 17th power?

8. Let positive integers K and d be given. Prove that there exists a positive integer n and a sequence of K positive integers $b_1, b_2, \dots, b_K$ such that the number n is a d -digit palindrome in all number bases $b_1, b_2, \dots, b_K$.
9. Let $S(k)$ denote the sum of digits of a positive integer k . Prove that there exists $k \in \mathbb{N}$ that does not contain the digit 9 in its decimal notation, such that $S(2^{2^{2017}} k) = S(k)$.
10. Let the function mirror map the digits 0, 1, 2, 5, 8 to 0, 1, 5, 2, 8, respectively. A positive integer $n = \overline{t_k t_{k-1} \dots t_1 t_0}$ is called *mirrorable* if all of its digits belong to the set $\{0, 1, 2, 5, 8\}$ with $t_0 \neq 0$, and we define $\text{mirror}(n) = \overline{\text{mirror}(t_0) \text{mirror}(t_1) \dots \text{mirror}(t_{k-1}) \text{mirror}(t_k)}$ (in other words, the function mirror stands for the mirror image of a calculator display). Find all positive integers n with the following properties:
- 1° n is mirrorable and $\text{mirror}(n) = n$;
 - 2° n^2 is mirrorable and $\text{mirror}(n^2) = n^2$.
11. Let the function rot map the digits 0, 1, 2, 5, 6, 8, 9 to 0, 1, 2, 5, 9, 8, 6, respectively. A positive integer $n = \overline{t_k t_{k-1} \dots t_1 t_0}$ is called *rotatable* if all of its digits belong to the set $\{0, 1, 2, 5, 6, 8, 9\}$ with $t_0 \neq 0$, and we define $\text{rot}(n) = \overline{\text{rot}(t_0) \text{rot}(t_1) \dots \text{rot}(t_{k-1}) \text{rot}(t_k)}$ (in other words, the function rot stands for rotating the display of a calculator for 180°). Prove that there exist infinitely many positive integers n with the following properties:
- 1° n is rotatable and $\text{rot}(n) = n$;
 - 2° n^2 is rotatable and $\text{rot}(n^2) = n^2$;
 - 3° $41 \mid n$.
12. Prove that for each $K \in \mathbb{N}$ there exists a number base b and K triples of Fibonacci numbers (F_u, F_v, F_w) such that, when they are written in the base b , their concatenation $\overline{F_u F_v F_w}$ is also a Fibonacci number written in the base b .
13. Prove that, in the decimal representation of $\sqrt[3]{3}$, there is a digit different from 2 on some position between 1 000 000 and 3 141 592 behind the decimal point.
14. Prove that there are only finitely many powers of 2 whose sum of digits is 2024.
15. The product of the binomial coefficient $\binom{64}{21}$ and an unknown odd positive integer equals

$$5*6*0*8*862*1*7*7*4*4*512*9**.$$

Find the missing digits.